

ORIGINAL GOALS Aspirational Design Spec (NOT current state)

*[!] READ THIS FIRST. This document is the **original, aspirational design target for the trading system the goals *as first specified*. It is NOT a description of what is currently built.** Many items below are aspirational, not yet implemented, infeasible with available data/infrastructure, or were later found by testing to be counterproductive.*

For the honest status of what is actually implemented vs. not, see GOAL_IMPLEMENTATION_AUDIT.md. For what the built system actually achieved, see BACKTEST_SUMMARY.md.

*Purpose of this doc: to gather outside opinions on the *original vision*.*

System concept

An autonomous quantitative statistical decision engine (not a predictor) that allocates capital across stocks and crypto using probabilistic expected value (EV), regime detection, and adaptive learning.

Core objective: maximize risk-adjusted returns (Sharpe, Sortino) while minimizing drawdowns, tail risk, overfitting, liquidity failure, regime collapse.

Primary rule: capital preservation > returns > growth.

Capital structure (dynamic)

- 0-100% Cash (default safe state; never required to be fully invested)
- 0-70% Stocks
- 0-30% Crypto

Universe

- Stocks: 25-40 liquid equities/ETFs. Core: SPY, QQQ, AAPL, MSFT, NVDA, AMZN, META, TSLA. Secondary: sector ETFs + mid caps.
- Crypto: 5-12 assets. Core: BTC, ETH. Secondary: large-cap alts only.
- Microcap rule (strict, experimental only): Stocks max 10% / 1% per position; Crypto max 5% / 0.5% per token.

Strategy structure

- Long-term (70-80%): trend following, factor investing, macro-aligned
- Medium-term (15-25%): swing trading, momentum cycles, mean reversion
- Short-term (5-10%): intraday / volatility, event-driven

Expected-value (EV) engine

$EV = P(\text{win}) \times \text{Avg Win} - P(\text{loss}) \times \text{Avg Loss}$.

Trade only if $EV > 0$ after fees, slippage, and risk adjustment.

Regime detection

Bull Trend - Bear Trend - Sideways/Chop - High Volatility - Crisis - Recovery.

Strategy weights adjust dynamically per regime.

No-trade conditions -> DEFAULT = CASH

EV $\leq$ 0 - insufficient liquidity - spread too wide - confidence < 60% - regime conflict - risk budget exhausted - correlation overload.

Position sizing (risk-capped Kelly)

Fractional Kelly, max 25%. Hard caps: Stocks 5-8%, Crypto 8-12% per position. Adjust by volatility regime.

Risk management (hard rules)

Max portfolio drawdown 15-25% - daily loss stop 3-5% - **3 consecutive loss days -> freeze trading** - no averaging down unless EV improves.

Cash rule

Cash is a valid position; may hold 0-100% cash indefinitely if EV insufficient, regimes unclear, or risk elevated.

Biweekly capital (e.g., +\$100 biweekly)

Deposits are NOT auto-invested held in cash until EV thresholds met. Max idle: 90 days.

Data sources & initial weighting (self-adjusting)

- Market + Price Data: 30%
- Filings / Insider (SEC EDGAR 10-K/Q/8-K, Form 4, 13F/13D/13G, congress): 25%
- Alternative data (Reddit/GitHub/etc.): 20%
- Macro (FRED/BLS/Fed, CFTC COT): 10%
- News (Reuters/Bloomberg/WSJ/FT): 10%
- Messaging apps (Discord/Telegram): 5%

Weights self-adjust based on predictive performance.

Learning loop

Per signal track: accuracy - Sharpe contribution - drawdown impact - decay rate. Increase weight if predictive; decrease if noisy.

Portfolio structure

- Core (95%): stable, high-liquidity, long-term
- Experimental (5%): microcaps, emerging narratives, high-variance
- Shadow (unlimited paper): all new ideas tested here first

Narrative system

Track lifecycle: emergence -> acceleration -> peak -> decline. Avoid crowded late-stage narratives.

Futures rule

Not in v1. Allowed only after 12+ months successful paper trading, a stable equity curve, and controlled drawdowns.

Decision hierarchy

1. Regime detection -> 2. EV calculation -> 3. Risk validation -> 4. Correlation check -> 5. Position sizing (Kelly capped) -> 6. Execution decision.

Required output (every decision)

Asset - Action (BUY / HOLD / SELL / NO TRADE) - EV score - Risk score -
Confidence - Position size - Reasoning - Data sources used.

Final intended behavior

Adaptive (not binary) - probabilistic (not absolute) - continuously active (not stalled) - risk-aware (not overly restrictive). **Maximize risk-adjusted returns while preserving capital not minimize activity.**

Reminder: the above is the ORIGINAL VISION for feedback. It is not a claim about what currently exists. See GOAL_IMPLEMENTATION_AUDIT.md for actual status.